

ML LAB EXPT – 5

Aim of the Experiment:

Develop a program to implement k-Nearest Neighbour algorithm to classify the randomly generated 100 values of x in the range of [0,1]. Perform the following based on dataset generated.

- Label the first 50 points {x₁,.....,x₅₀} as follows: if (x_i ≤ 0.5), then x_i ∈ Class1, else x_i ∈ Class2**
- Classify the remaining points, x₅₁,.....,x₁₀₀ using KNN. Perform this for k=1,2,3,4,5,20,30**

```
import numpy as np
import matplotlib.pyplot as plt
from collections import Counter

def knn_classify(k, x, points, labels):
    distances = [np.abs(x - point) for point in points]
    k_nearest_labels = [labels[i] for i in
                        np.argsort(distances)[:k]]
    most_common = Counter(k_nearest_labels).most_common(1)
    return most_common[0][0]

# Generate 100 random x values between 0 and 1
x_values = np.random.rand(100)

# Label the first 50 points
labels = []
for i in range(100):
    if i < 50:
        if x_values[i] <= 0.5:
            labels.append("Class1")
        else:
            labels.append("Class2")
    else:
        labels.append(None) # Initialize labels for the remaining
                           #points to None

# Classify the remaining 50 points using k-NN for different values
of k
k_values = [1, 2, 3, 4, 5, 20, 30]
for k in k_values:
    for i in range(50, 100):
        labels[i] = knn_classify(k, x_values[i], x_values[:50],
                                labels[:50])
```

```
# Visualization for each k
plt.figure(figsize=(8, 6))
plt.scatter(x_values[:50], [0] * 50, c=['red' if label ==
    'Class1' else 'blue' for label in labels[:50]],
    label='Training Data')
plt.scatter(x_values[50:], [0] * 50, c=['green' if label ==
    'Class1' else 'purple' for label in labels[50:]],
    label=f'k = {k}')
plt.xlabel('x values')
plt.ylabel('Class')
plt.title(f'k-NN Classification (k={k})')
plt.legend()
plt.show()
```

Output:







